Niklas Erdmann

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About Me: Personal curiosity, peers, and experiences during my prior education led me to be interested in researching applications of Artificial Intelligence (AI) in our world's problems and to pursue a PhD in Oslo. My experiences, interests, and collaborations enabled me to many cross-disciplinary interactions in which I enjoy contributing with my broad background in AI, Statistics, and Psychology. Continuous passion for research and teaching guides me towards more academic endeavours.

Post-Graduate Education:

PhD., Dep. of Technology Systems, University of Oslo **03.2023 - 03.2027**
Title: *Artificial Intelligence-based Prognosis of Dynamic Energy Systems*

PhD project as a collaboration between the University of Oslo, the Norwegian Energy Research Institute, and the Norwegian Defense Research Establishment around the forecasting of intelligent dynamic energy systems. A main focus is placed on probabilistic forecasting of solar irradiance as an application domain.

Publications

Erdmann, N., Bentsen, L., Stenbro, R., Riise, H. N., Warakagoda, N. D., & Engelstad, P. E. (2025). Multi-Horizon Time Series Forecasting of Non-Parametric CDFs with Deep Lattice Networks. In *Proceedings of the 40th Annual AAAI Conference on Artificial Intelligence.* (arXiv, Code and Data)

Eriksen, E. W., Nygard, M. M., Erdmann, N., & Riise, H. N. (2025). Deep Learning Multi-Horizon Irradiance Nowcasting: A Comparative Evaluation of Three Methods for Leveraging Sky Images. *IEEE Journal of Photovoltaics*. [Manuscript submitted for publication]

Erdmann, N., Bentsen, L. Ø., Stenbro, R., Riise, H. N., Warakagoda, N., & Engelstad, P. (2024). Deep and Probabilistic Solar Irradiance Forecast at the Arctic Circle. In *Proceedings of the 2024 IEEE 52nd Photovoltaic Specialist Conference (PVSC)* (pp. 1599–1606). (IEEE, arXiv)

Zieglmeier, S., Erdmann, N., & Warakagoda, N. D. (2025). Reinforcement Learning for Pollution Detection in a Randomized, Sparse and Nonstationary Environment with an Autonomous Underwater Vehicle. *arXiv preprint arXiv:2510.26347*. (arXiv)

Education:

MSc. Artificial Intelligence, University of Groningen **09.2019 - 07.2022**
English-taught, Grade: 8.4/10, ECTS: 121
Master Thesis: *"Dynamic Modeling of a P(VDF-TrFE-CTFE)-Composite Soft Actuator"*.(RUG)

Completed courses across the AI spectrum: Data Science, Machine/Deep Learning, Computer Vision, Pattern Recognition, Multi-Agent Systems, Reinforcement Learning, and Neuroscience.

Publication

D'Anniballe, R., Erdmann, N., Selleri, G., & Carloni, R. (2022). Dynamic Modeling of P (VDF-TrFE-CTFE)-based Soft Actuators via Echo State Networks. In *Proceedings of the 2022 IEEE/ASME International Conference on advanced intelligent mechatronics (aim)* (pp. 118–124). (IEEE, RUG)

Pre-Master Artificial Intelligence, University of Groningen **09.2018 - 08.2019**
English-taught, Grade: 8.1/10, ECTS: 50
Selection of Bachelor's courses targeted at providing the foundation for the AI Master's (including Calculus, Programming, and Machine Learning).

Spring School Cognitive Modeling, University of Groningen **03.2018**
Introduction to several computational and statistical paradigms utilized for cognitive modeling (i.e. PRIMs, ACT-R, Nengo)

BSc. Psychology, University of Groningen **09.2015 - 01.2019**
English-taught, Grade: 7.6/10, ECTS: 200 Honours Bachelor Thesis: *"The Effect of Context Manipulation on Recall"*.

Completion of an additional curriculum as part of the Excellence Programme for talented and ambitious students. General focus towards Cognitive Psychology, Statistics, and Research Methods, with e.g. a minor in Artificial Intelligence focusing on Neurophysics and Cognitive Modeling.

Teaching Experience:

PhD. Teaching Component, University of Oslo **09.2024 - 03.2027**
1 out of 4 years of my PhD is contractually spent on teaching activities, including teaching education. Activities consist of holding lectures, designing assignments, and assisting with exams. At this moment, this includes independent supervision of 5 Master's thesis projects.

Teaching Assistant, University of Groningen **09.2019 - 07.2021**
This included leading tutorial sessions, grading assignments and exams, and geading and coordinating a group of other TAs.

Other Activities:

University Board Representation

Representative for PhDs and PostDocs at the Department Board (Dep. of Technology Systems, University of Oslo) (2025).
Vice-representative for PhDs and PostDocs at the Faculty Board (Faculty of Mathematics and Natural Sciences, University of Oslo) (2024).

Awards

Best Poster Award, with "Deep and probabilistic solar irradiance forecast at the arctic circle." during the 2024 IEEE 52nd Photovoltaic Specialist Conference (PVSC).
Best Paper Award, with "Reinforcement learn- ing for pollution detection in a randomized, sparse and nonstationary environ- ment with an autonomous underwater vehicle." during the educational component of my PhD in Oslo.
Best Performer: Leukemia Challenge, successfully classifying a set of biomarkers for Leukemia during my Master's in Groningen.

Relevant Skills and Experiences:

Languages
Full English education since my Bachelor's and intensive introductory courses to third languages.

German (Native), English (Professional), Dutch & Norwegian (Elementary).

Programming
Regular programming experience and education since my Pre-Master's.

Language familiarity with Python, C, R, and Matlab.
Python packages: TensorFlow, PyTorch, Scipy, Optuna and OpenCV.
Other tools: Git, LaTex, Slurm